

TASK-3

Login System with Validation

Aim

To develop a secure login system that validates user inputs using JavaScript, verifies credentials from a database, and dynamically displays error messages.

Algorithm

1. Start the process.
2. Create a login form with username and password fields.
3. Capture user input using JavaScript.
4. Validate inputs (check empty fields and format).
5. If validation fails then display error message.
6. If validation passes, then send data to server.
7. Connect to database.
8. Check whether username and password match records.
9. If matched then login successful and redirect.
10. If not matched then show invalid credentials message.
11. Stop the process.

Program

Login.html :

```
<!DOCTYPE html>

<html>

<head>

<title>Login Page</title>

<script src="script.js"></script>

</head>
```

```
<body>

<h2>Login Form</h2>

<form onsubmit="return validateForm()" action="login.php" method="POST">

Username: <input type="text" id="username" name="username"><br><br>

Password: <input type="password" id="password" name="password"><br><br>

<span id="error" style="color:red;"></span><br>

<input type="submit" value="Login">

</form>

</body>

</html>
```

Script.js:

```
function validateForm() {

let username = document.getElementById("username").value;

let password = document.getElementById("password").value;

let error = document.getElementById("error");

if (username === "" || password === "") {

error.innerHTML = "All fields are required!";

return false;

}

if (password.length < 6) {

error.innerHTML = "Password must be at least 6 characters!";
```

```
return false;
```

```
}
```

```
return true;
```

```
}
```

Login.php:

```
<?php
```

```
$conn = new mysqli("localhost", "root", "", "test_db");
```

```
if ($conn->connect_error) {
```

```
    die("Connection failed: " . $conn->connect_error);
```

```
}
```

```
$username = $_POST['username'];
```

```
$password = $_POST['password'];
```

```
$sql = "SELECT * FROM users WHERE username='$username' AND  
password='$password'";
```

```
$result = $conn->query($sql);
```

```
if ($result->num_rows > 0) {
```

```
    echo "Login Successful!";
```

```
} else {
```

```
        echo "Invalid Username or Password!";  
    }  
}
```

```
$conn->close();
```

```
?>
```

SQL Table:

```
CREATE DATABASE test_db;
```

```
CREATE TABLE users (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    username VARCHAR(50),  
    password VARCHAR(50)  
);
```

OUTPUT:



Login Form

Username:

Password:



Result

Thus, the login page validates inputs using JavaScript, displays dynamic error messages for empty or incorrect inputs, and allows access only for valid credentials while rejecting invalid ones.